Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 8 · SLO 1 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 8 : SLO â€“ 1 | Asymptotic Notations â€” Big O, Big Omega, Big Theta

I. Fill in the Blanks

1. Big O â€” the upper bound on running time.
2. Big Omega (Î©) â€” the lower bound.
3. Big Theta (Î˜) â€” the tight bound.
4. worst â€” Big O analyzes worst-case performance.
5. best â€” Big Omega analyzes best-case performance.
6. upper and lower â€” Î˜ bounds from both sides.
7. O(n) â€” linear search.
8. O(log n) â€” binary search.
9. O(nÂ²) â€” bubble sort worst case.
10. O â€” f(n) = 3n + 2 is O(n).
11. O â€” f(n) = 5nÂ² + 3n + 7 is O(nÂ²).
12. lower-order â€” constants and lower-order terms are ignored.
13. O(1) (constant) â€” same time regardless of input size.
14. Î© â€” f(n) â‰¥ cÂ·g(n) for n â‰¥ nâ‚€ means f(n) = Î©(g(n)).
15. O and Î© â€” Î˜(g(n)) means both O(g(n)) and Î©(g(n)).